

Lecture 20

Short Run Costs and Profit Maximization

ECO303

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Quantitative Size of Shifts in Costs Curves

- The increase in costs will be largely influenced by
 - The relative significance of the input in the production process
 - The ability of firms to substitute another input for the one that has risen in price

EXAMPLE 10.3 Shifting the Cobb–Douglas Cost Function

- Cobb-Douglas cost function:

$$C(v, w, q) = vk + wl = q^{1/\alpha+\beta} B v^{\alpha/\alpha+\beta} w^{\beta/\alpha+\beta}$$

where $B = (\alpha + \beta) \alpha^{-\alpha/\alpha+\beta} \beta^{-\beta/\alpha+\beta}$

- Assume $\alpha = \beta = 0.5$, the total cost curve is greatly simplified

$$C(v, w, q) = vk + wl = 2qv^{0.5}w^{0.5}$$

EXAMPLE 10.3 Shifting the Cobb–Douglas Cost Function

- If $v = 3$ and $w = 12$, the relationship is

$$C(3,12,q) = 2q\sqrt{36} = 12q$$

- $C = 480$ to produce $q = 40$
- $AC = C/q = 12$
- $MC = \partial C/\partial q = 12$

- If $v = 3$ and $w = 27$, the relationship is

$$C(3,27,q) = 2q\sqrt{81} = 18q$$

- $C = 720$ to produce $q = 40$
- $AC = C/q = 18$
- $MC = \partial C/\partial q = 18$

Contingent Demand for Inputs

- Contingent demand functions
 - For all of the firms inputs can be derived from the cost function
- Shephard's lemma
 - The contingent demand function for any input is given by the partial derivative of the total-cost function with respect to that input's price

Contingent Demand for Inputs

- Shepherd's lemma
 - Is one result of the envelope theorem
 - The change in the optimal value
 - In a constrained optimization problem
 - With respect to one of the parameters
 - Can be found by differentiating the Lagrangian with respect to the changing parameter

EXAMPLE 10.4 Contingent Input Demand Functions

- Fixed proportions, $C(v, w, q) = q(v/\alpha + w/\beta)$
 - Contingent demand functions are quite simple:

$$k^c(v, w, q) = \frac{\partial C(v, w, q)}{\partial v} = \frac{q}{\alpha}$$

$$l^c(v, w, q) = \frac{\partial C(v, w, q)}{\partial w} = \frac{q}{\beta}$$

Short-Run, Long-Run Distinction

- In the short run
 - Economic actors have only limited flexibility in their actions
- Assume
 - The capital input is held constant at k_1
 - The firm is free to vary only its labor input
- The production function becomes

$$q = f(k_1, l)$$

Short-Run Total Costs

- Short-run total cost for the firm is

$$SC = vk_1 + wl$$

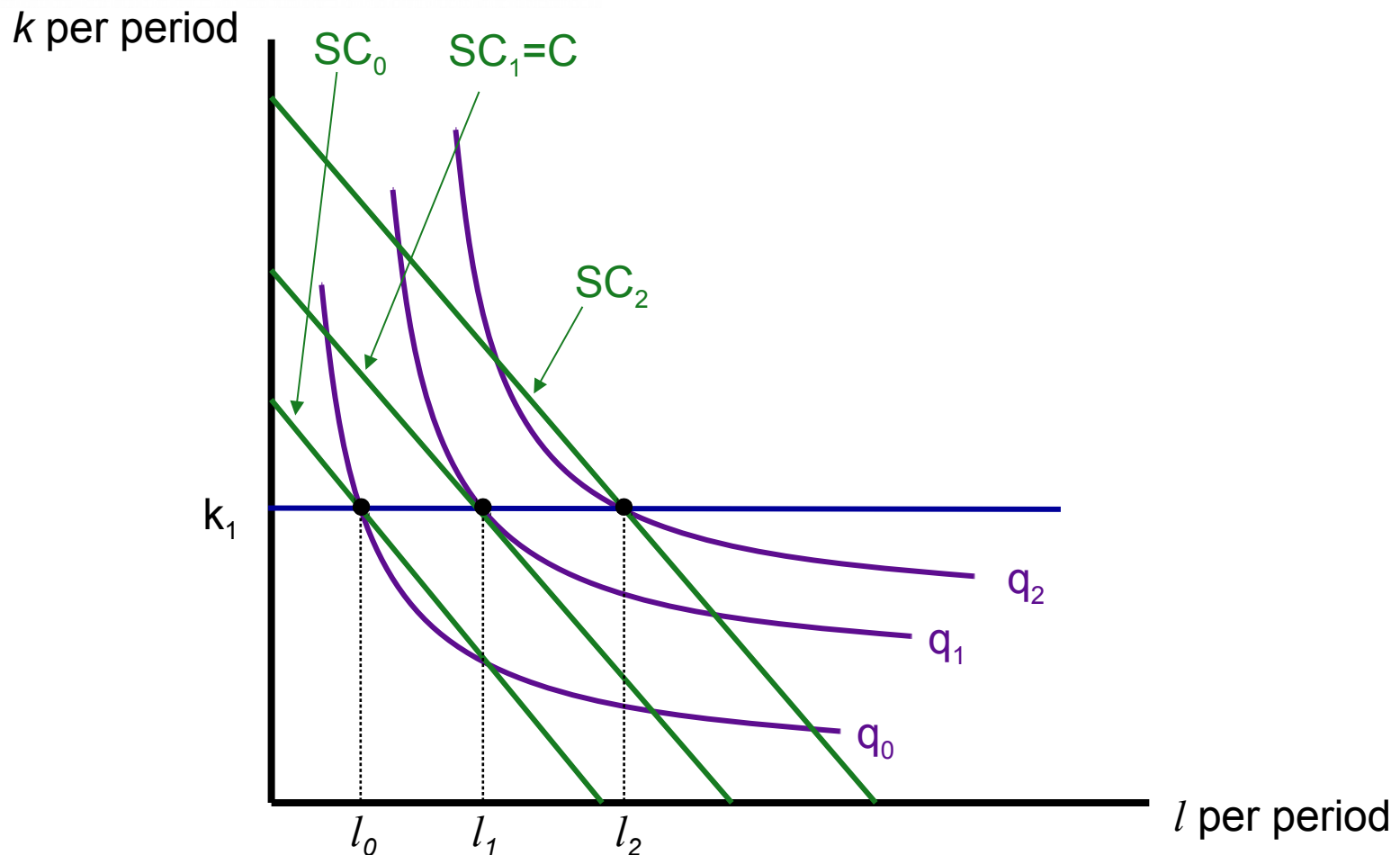
- There are two types of short-run costs:
 - Short-run fixed costs are costs associated with fixed inputs (vk_1)
 - Short-run variable costs are costs associated with variable inputs (wl)

Short-Run Total Costs

- Short-run costs
 - Are not minimal costs for producing the various output levels
 - The firm does not have the flexibility of input choice
 - To vary its output in the short run, the firm must use nonoptimal input combinations
 - The *RTS* will not be equal to the ratio of input prices

FIGURE 10.7

“Nonoptimal” Input Choices Must Be Made in the Short Run



Because capital input is fixed at k , in the short run the firm cannot bring its *RTS* into equality with the ratio of input prices. Given the input prices, q_0 should be produced with more labor and less capital than it will be in the short run, whereas q_2 should be produced with more capital and less labor than it will be.

Short-Run Marginal and Average Costs

- The short-run average total cost (SAC) function is

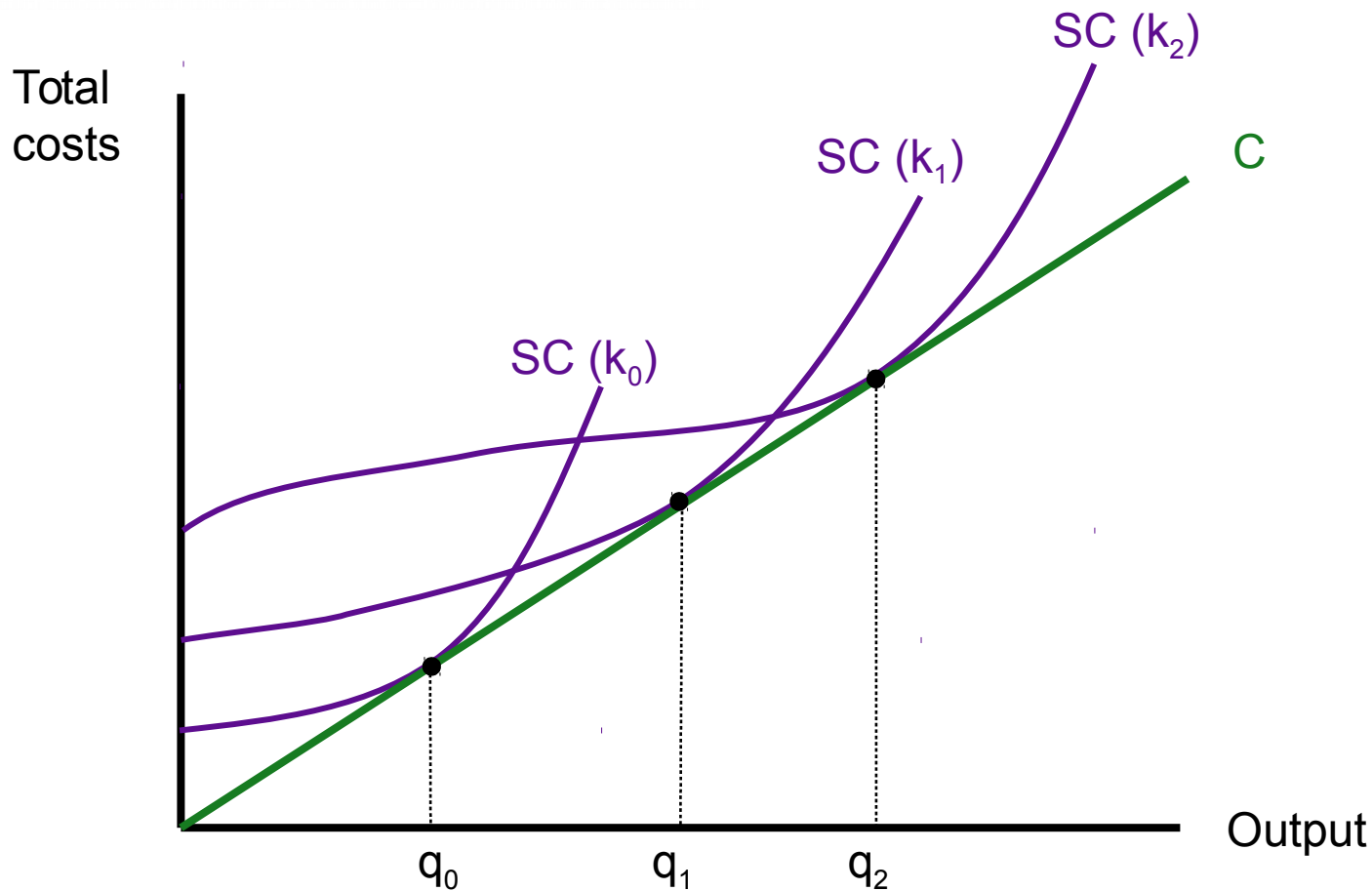
$$SAC = \text{total costs/total output} = SC/q$$

- The short-run marginal cost (SMC) function is

$$SMC = \text{change in } SC/\text{change in output} = \partial SC/\partial q$$

FIGURE 10.8 (a) Constant returns to scale

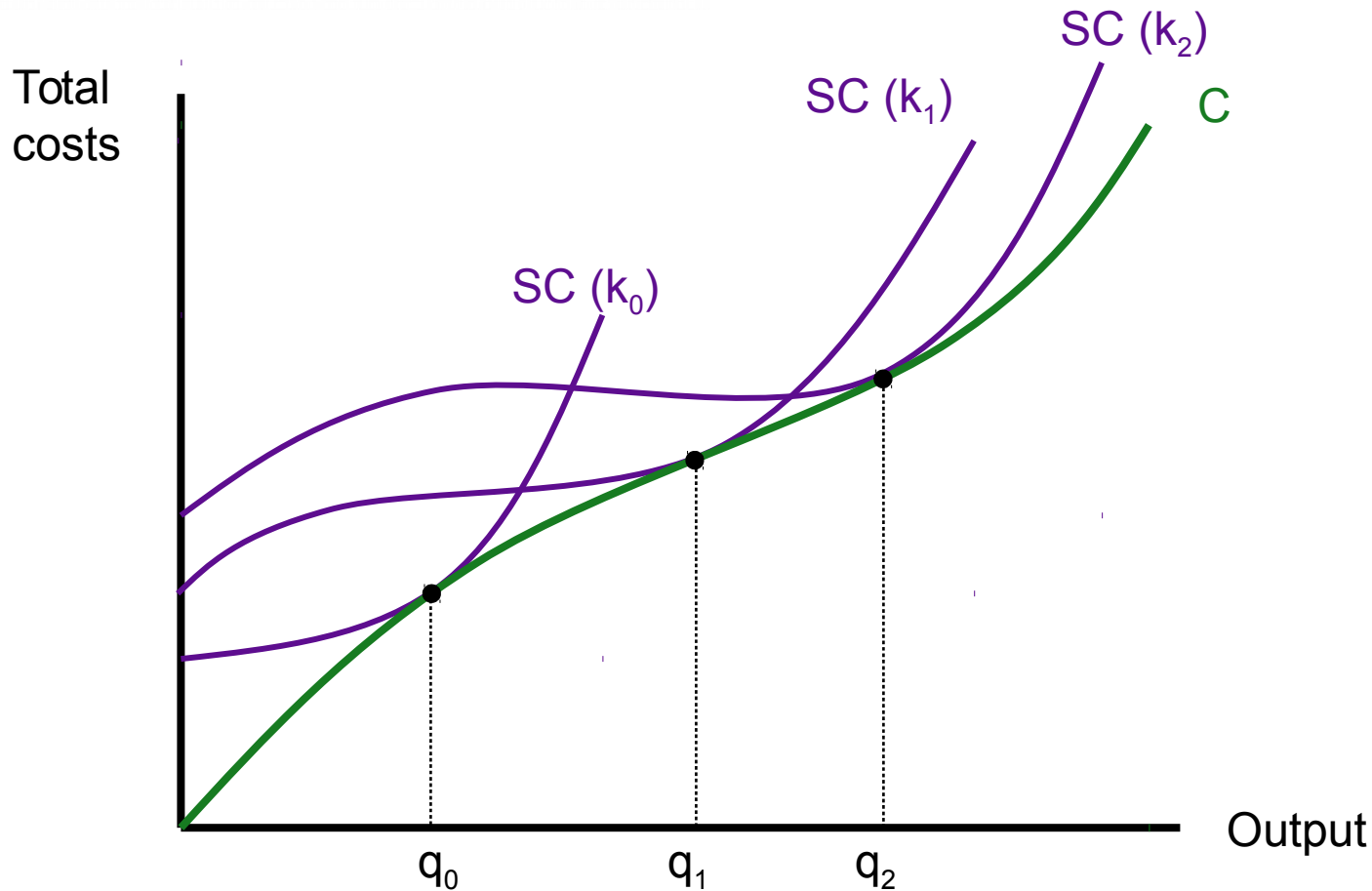
Two Possible Shapes for Long-Run Total Cost Curves



By considering all possible levels of capital input, the long-run total cost curve (C) can be traced. In (a), the underlying production function exhibits constant returns to scale: In the long run, although not in the short run, total costs are proportional to output.

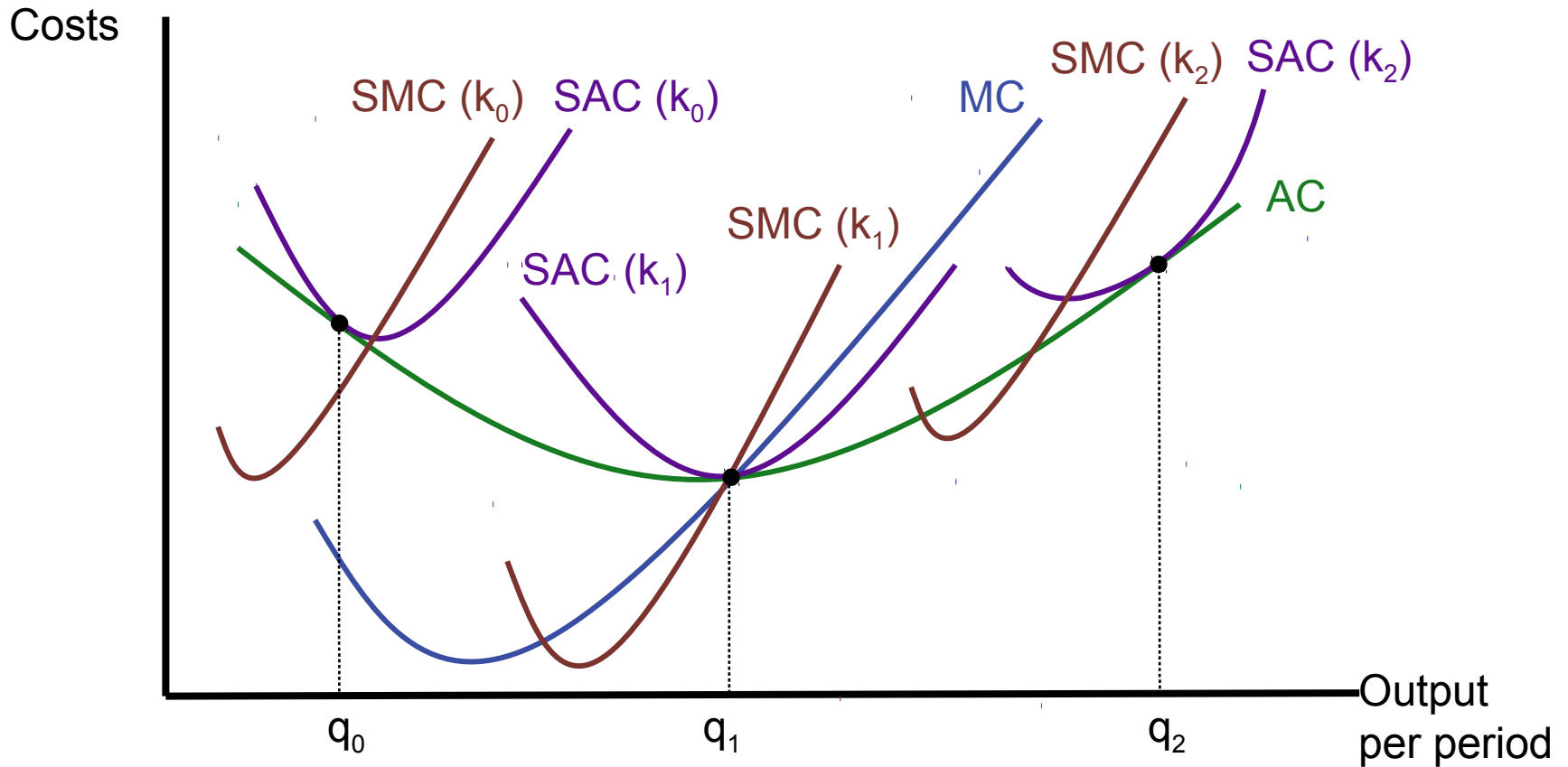
FIGURE 10.8 (b) Cubic total cost curve case

Two Possible Shapes for Long-Run Total Cost Curves



By considering all possible levels of capital input, the long-run total cost curve (C) can be traced. In (b), the long-run total cost curve has a cubic shape, as do the short-run curves. Diminishing returns set in more sharply for the short-run curves, however, because of the assumed fixed level of capital input.

Average and Marginal Cost Curves for the Cubic Cost Curve Case



This set of curves is derived from the total cost curves shown in Figure 10.8. The AC and MC curves have the usual U-shapes, as do the short-run curves. At q_1 , long-run average costs are minimized. The configuration of curves at this minimum point is important.

Short-Run and Long-Run Costs

- At the minimum point of the AC curve:
 - The MC curve crosses the AC curve
 - $MC = AC$ at this point
 - The SAC curve is tangent to the AC curve
 - SAC (for this level of k) is minimized at the same level of output as AC
 - SMC intersects SAC also at this point

$$AC = MC = SAC = SMC$$

The Nature and Behavior of Firms

- A firm
 - An association of individuals
 - Who have organized themselves for the purpose of turning inputs into outputs
- Firms
 - All shapes and sizes
 - Structure of their management
 - Financial instruments

The Nature and Behavior of Firms

- Simple model of a firm
 - Technology given by the production function, $f(k, l)$
 - Inputs: labor (l) and capital (k)
 - Run by an entrepreneur
 - Makes all the decisions
 - Receives all the profits and losses from the firm's operations
 - Acts in his or her own self-interest
 - Maximize the firm's profits

The Nature and Behavior of Firms

- Complicating factors
 - Management team
 - Shareholders
 - Owners
 - Outsourcing
- Firms
 - Are not individuals but can be much more complicated organizations
 - Firm's objectives: profit maximization

Profit Maximization

- A profit-maximizing firm
 - Chooses both its inputs and its outputs
 - With the sole goal of achieving maximum economic profits
 - Seeks to maximize the difference between total revenue and total economic costs
 - Make decisions in a “marginal” way
 - Examine the marginal profit obtainable from producing one more unit of hiring one additional laborer

Output Choice

- Total revenue for a firm, $R(q) = p(q) \cdot q$
- Economic costs incurred, $C(q)$
 - In the production of q
- Economic profits, π
 - The difference between total revenue and total costs

$$\pi(q) = R(q) - C(q) = p(q) \cdot q - C(q)$$

Output Choice

- The necessary condition
 - For choosing the level of q that maximizes profits
 - Setting the derivative of the π function with respect to q equal to zero

$$\frac{d\pi}{dq} = \pi'(q) = \frac{dR}{dq} - \frac{dC}{dq} = 0$$

$$\frac{dR}{dq} = \frac{dC}{dq}$$

Output Choice

- Marginal revenue, MR

- The change in total revenue R resulting from a change in output q

$$\text{Marginal revenue} = MR = dR/dq$$

- Profit maximization

- Maximize economic profits
- Choose output q^* at which

$$MR(q^*) = MC(q^*)$$
$$MR = \frac{dR}{dq} = \frac{dC}{dq} = MC$$

Second-Order Conditions

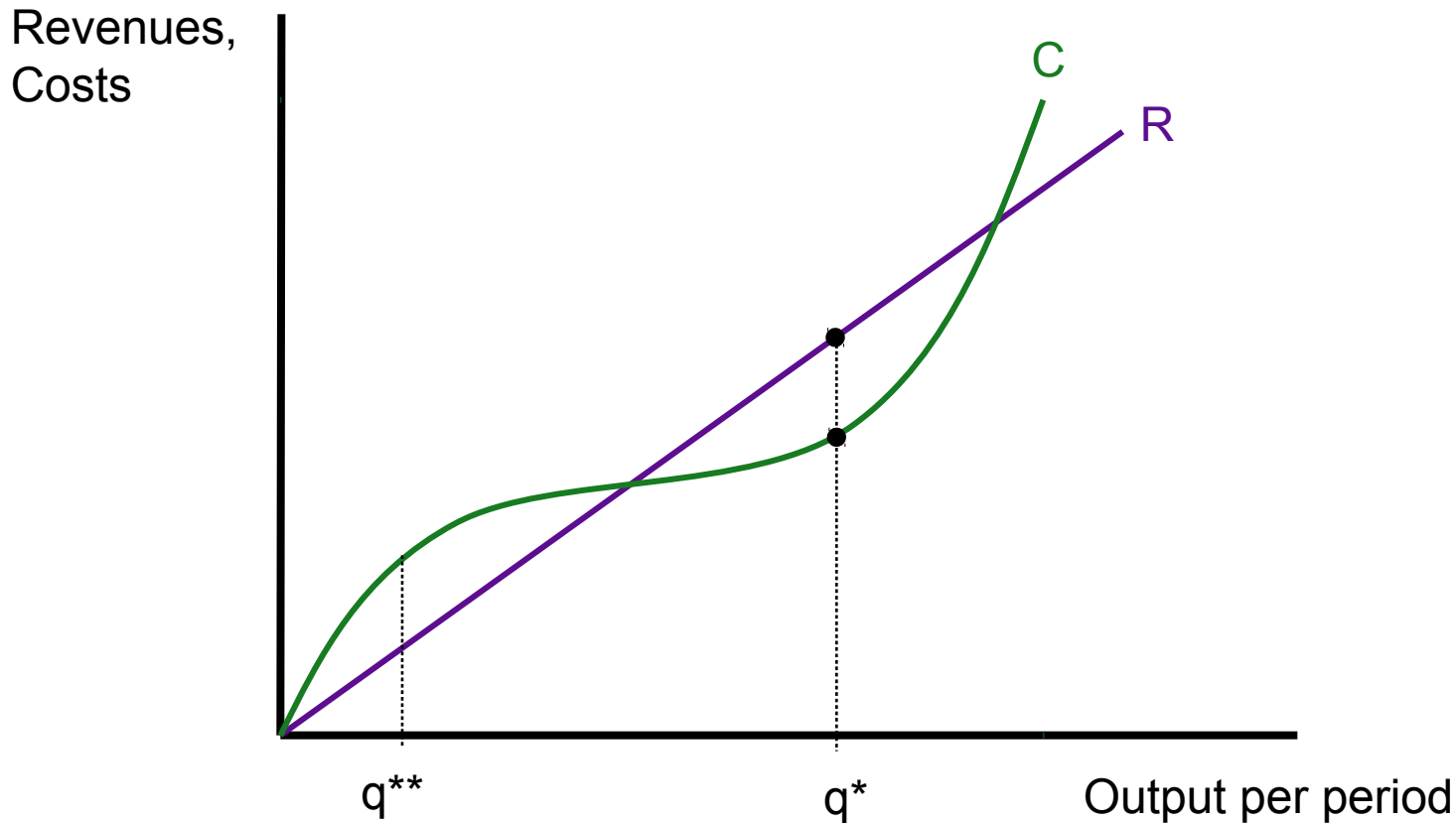
- $MR = MC$
 - Is only a necessary condition for profit maximization
- For sufficiency, it is also required:

$$\left. \frac{d^2 \pi}{dq^2} \right|_{q=q^*} = \left. \frac{d\pi'(q)}{dq} \right|_{q=q^*} < 0$$

- “marginal” profit must decrease at the optimal level of output, q^*
 - For $q < q^*$, $\pi'(q) > 0$
 - For $q > q^*$, $\pi'(q) < 0$

11.1 (a)

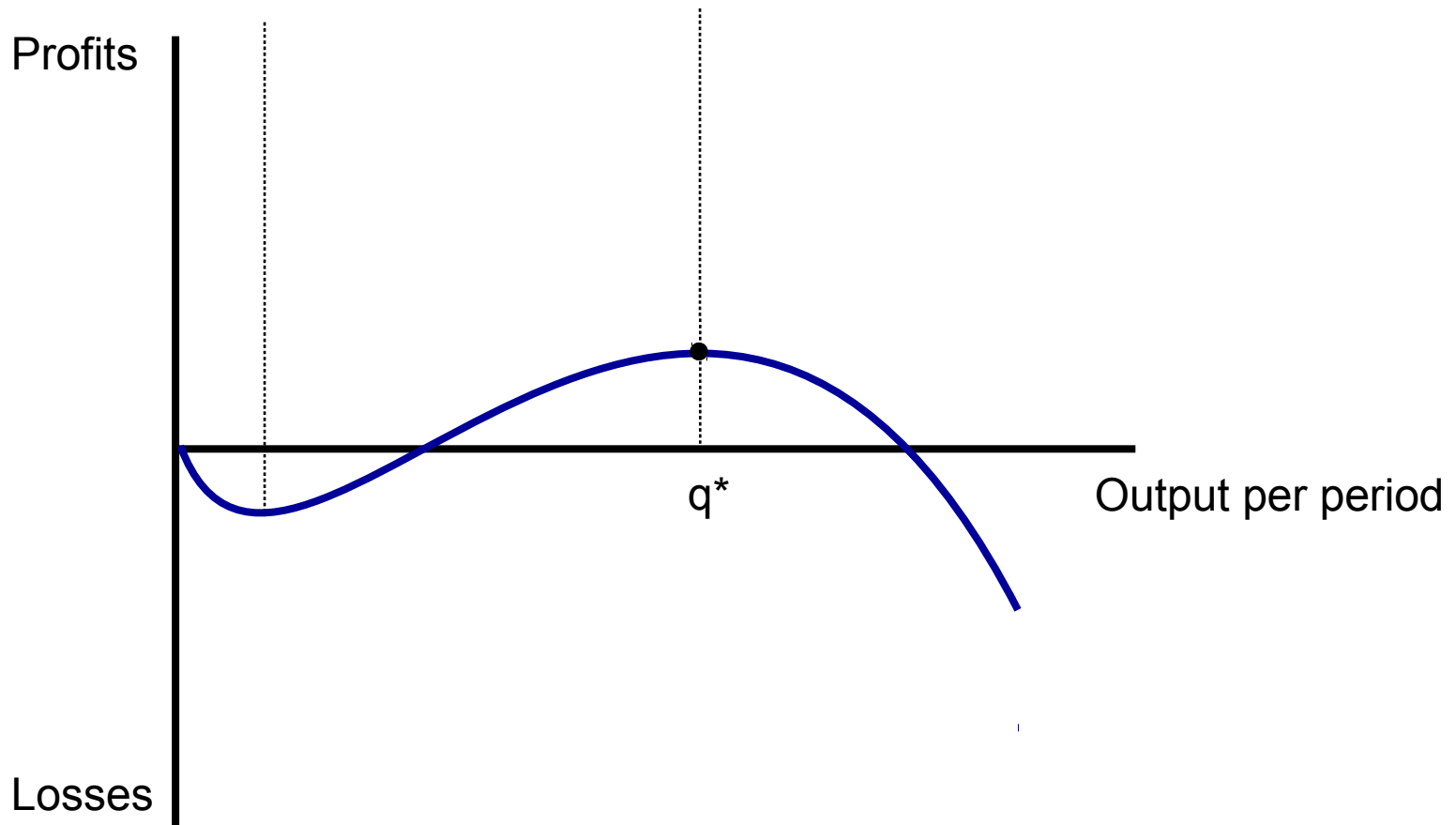
•Marginal Revenue Must Equal Marginal Cost for Profit Maximization



•Profits, defined as revenues (R) minus costs (C), reach a maximum when the slope of the revenue function (marginal revenue) is equal to the slope of the cost function (marginal cost). This equality is only a necessary condition for a maximum, as may be seen by comparing points q^* (a true maximum) and q^{**} (a local minimum), points at which marginal revenue equals marginal cost.

11.1 (b)

•Marginal Revenue Must Equal Marginal Cost for Profit Maximization



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Marginal Revenue

- Marginal revenue

- Equals price - If a firm can sell all it wishes without having any effect on market price
- If a firm faces a downward-sloping demand curve
 - More output can only be sold if the firm reduces the good's price

$$MR(q) = \frac{dR}{dq} = \frac{d[p(q) \cdot q]}{dq} = p + q \cdot \frac{dp}{dq}$$

Marginal Revenue

- Marginal revenue - is a function of output
 - If price does not change as quantity increases
 - $dp/dq = 0$, $MR = \text{price}$
 - The firm is a price taker
 - If price decreases as quantity increases
 - $dp/dq < 0$, $MR < \text{price}$

Marginal Revenue from a Linear Demand Function

- Demand curve for a sub sandwich is
$$q = 100 - 10p$$
- Solving for price: $p = -q/10 + 10$
- Total revenue: $R = pq = -q^2/10 + 10q$
- Marginal revenue: $MR = dR/dq = -q/5 + 10$
 - $MR < p$ for all values of q
- If the average and marginal costs are constant (\$4)
 - Profit maximizing quantity: $MR = MC$, so $q^*=30$
 - Price = \$7, and profits = \$90

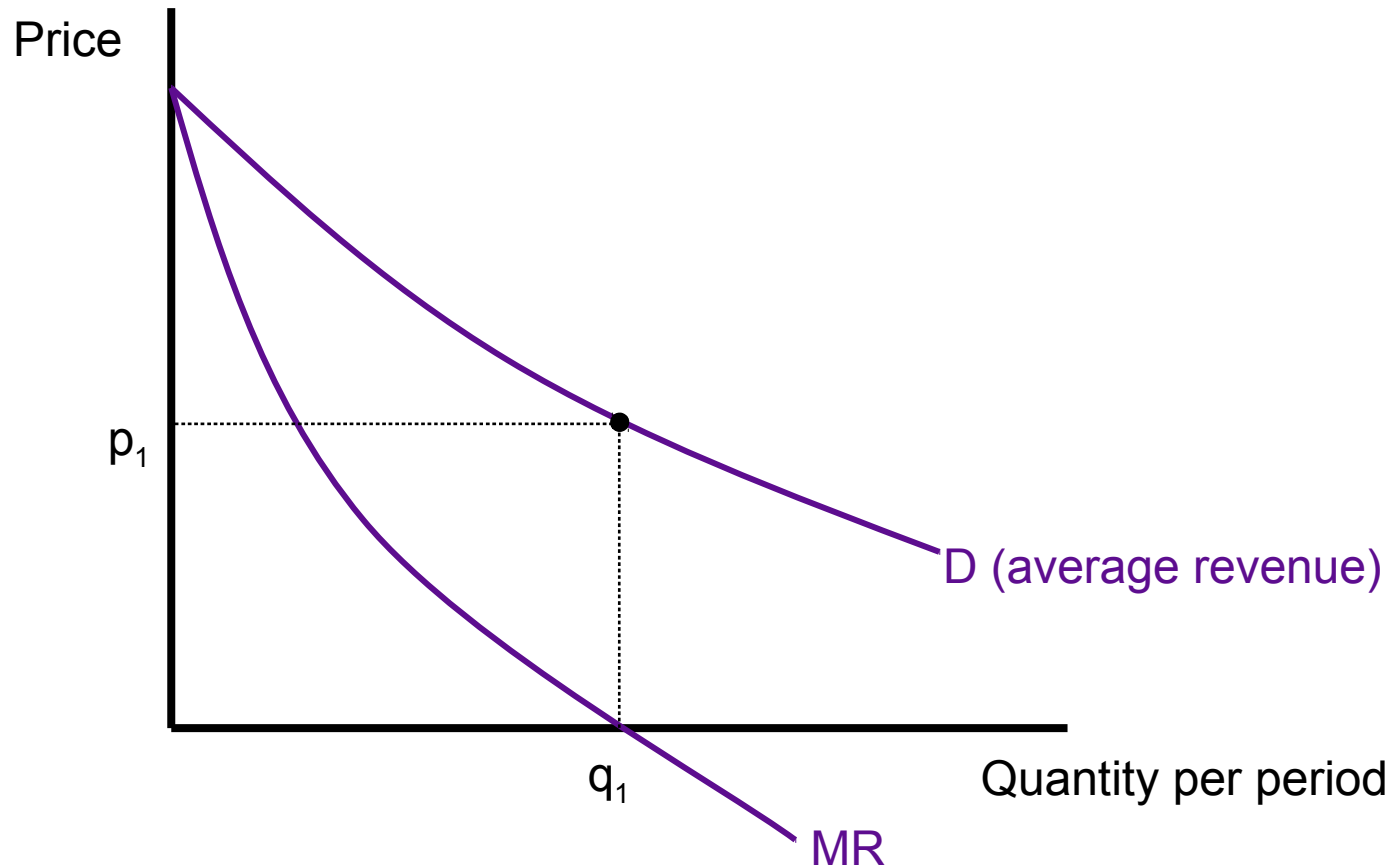
Average Revenue Curve

- Assume
 - That the firm must sell all its output at one price
 - So, we can think of the demand curve facing the firm as its average revenue curve
 - Shows the revenue per unit yielded by alternative output choices

Marginal Revenue Curve

- Marginal revenue curve
 - Shows the extra revenue provided by the last unit sold
 - Below the demand curve
 - In the case of a downward-sloping demand curve

•Market Demand Curve and Associated Marginal Revenue Curve



•Because the demand curve is negatively sloped, the marginal revenue curve will fall below the demand ("average revenue") curve. For output levels beyond q_1 , MR is negative. At q_1 , total revenues ($p_1 \cdot q_1$) are a maximum; beyond this point, additional increases in q cause total revenues to decrease because of the concomitant decreases in price.

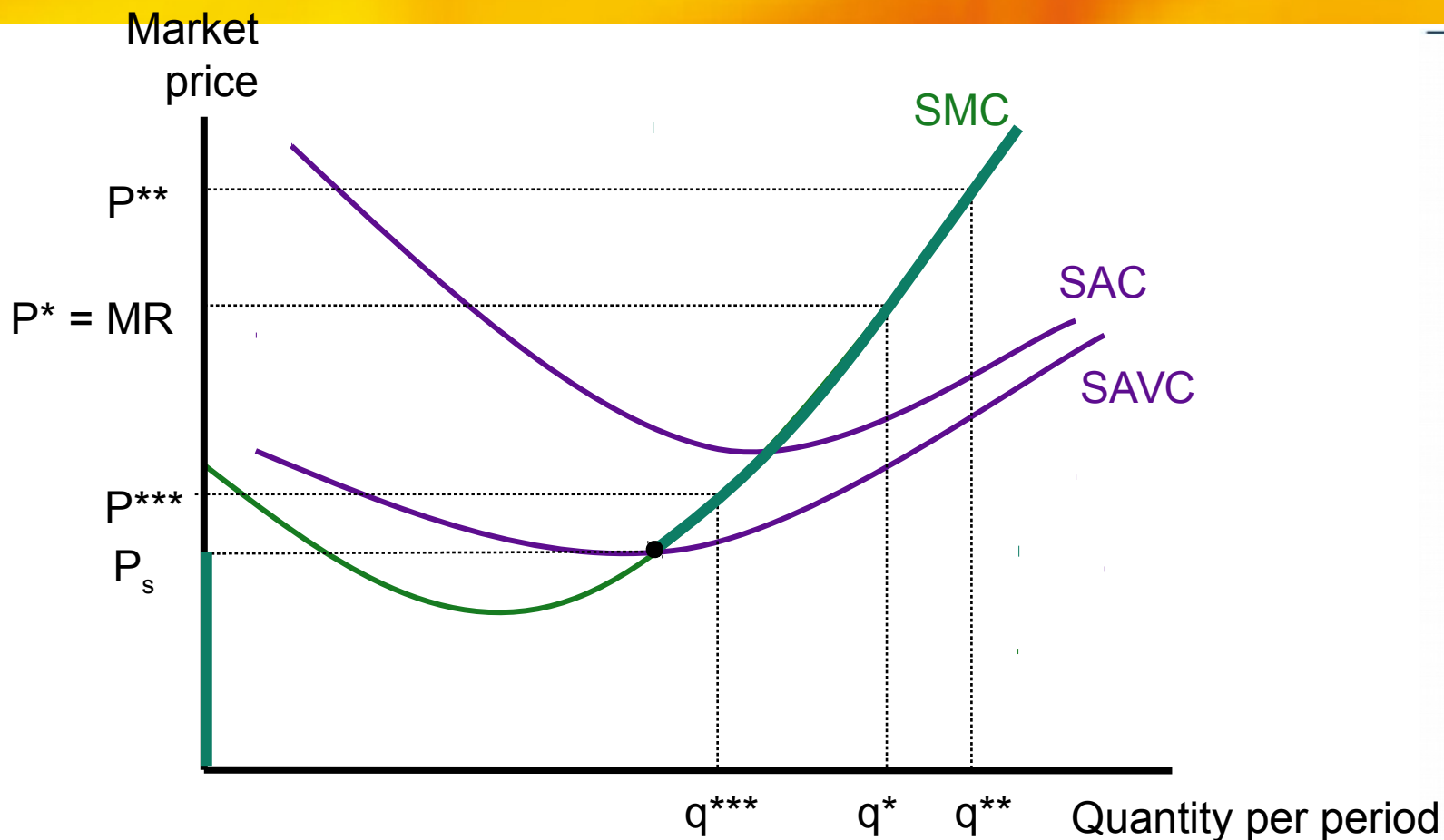
Marginal Revenue Curve

- When the demand curve shifts
 - The marginal revenue curve associated with it shifts as well
- A marginal revenue curve
 - Cannot be calculated without referring to a specific demand curve

Short-Run Supply by a Price-Taking Firm

- Price-taking firm
 - Demand curve facing the firm is a horizontal line through $P^*=MR$
 - Profit-maximizing output level, q^*
 - P^* = short run marginal costs
 - And marginal cost is increasing
 - At q^* , profits are
 - Positive if price $>$ average costs
 - Negative (loss) if price $<$ average costs
 - Zero if price = average costs

•Short-Run Supply Curve for a Price-Taking Firm



•In the short run, a price-taking firm will produce the level of output for which $SMC = P$. At P^* , for example, the firm will produce q^* . The SMC curve also shows what will be produced at other prices. For prices below $SAVC$, however, the firm will choose to produce no output. The heavy lines in the figure represent the firm's short-run supply curve.

Short-Run Supply by a Price-Taking Firm

- Short-run supply curve
 - The positively-sloped portion of the short-run marginal cost curve
 - Above the point of minimum average variable cost
 - It shows how much the firm will produce at every possible market price

Short-Run Supply by a Price-Taking Firm

- Firms will only operate in the short run
 - As long as total revenue covers variable cost
 - $p > SAVC$
- Firms will shut-down in the short-run
 - If $p < SAVC$
 - Produce no output

11.3 Short-Run Supply

- Firm's short-run total cost curve is

$$SC(v, w, q, k) = vk_1 + wq^{1/\beta}k_1^{-\alpha/\beta}$$

- Where k_1 is the level of capital held constant in the short run

- Short-run marginal cost is

$$SMC(v, w, q, k_1) = \frac{\partial SC}{\partial q} = \frac{w}{\beta} q^{(1-\beta)/\beta} k_1^{-\alpha/\beta}$$

- The price-taking firm will maximize profit where $p = SMC$

$$SMC = \frac{w}{\beta} q^{(1-\beta)/\beta} k_1^{-\alpha/\beta} = P$$

11.3 Short-Run Supply

- Quantity supplied will be

$$q = \left(\frac{w}{\beta} \right)^{-\beta/(1-\beta)} k_1^{-\alpha/(1-\beta)} P^{\beta/(1-\beta)}$$

- To find the firm's shut-down price, we need to solve for SAVC

$$SVC = wq^{1/\beta} k_1^{-\alpha/\beta}$$

$$SAVC = SVC/q = wq^{(1-\beta)/\beta} k_1^{-\alpha/\beta}$$

- $SAVC < SMC$ for all values of $\beta < 1$
 - There is no price low enough that the firm will want to shut down

Profit Functions

- A firm's economic profit can be expressed as a function of inputs

$$\pi = Pq - C(q) = Pf(k,l) - vk - wl$$

- Only k and l are under the firm's control
- P , v , and w - fixed parameters

- Profit function

- Shows its maximal profits as a function of the prices that the firm faces

$$\Pi(P, v, w) = \max_{k,l} \pi(k,l) = \max_{k,l} [Pf(k,l) - vk - wl]$$

Properties of the Profit Function

- Homogeneity
 - The profit function is homogeneous of degree one in all prices
- Nondecreasing in output price, P
- Nonincreasing in input prices, v , and w
- Convex in output prices

$$\frac{\Pi(P_1, v, w) + \Pi(P_2, v, w)}{2} \geq \Pi\left[\frac{P_1 + P_2}{2}, v, w\right]$$

Envelope Results

- Apply the envelope theorem
 - To see how profits respond to changes in output and input prices

$$\frac{\partial \Pi(P, v, w)}{\partial P} = q(P, v, w)$$

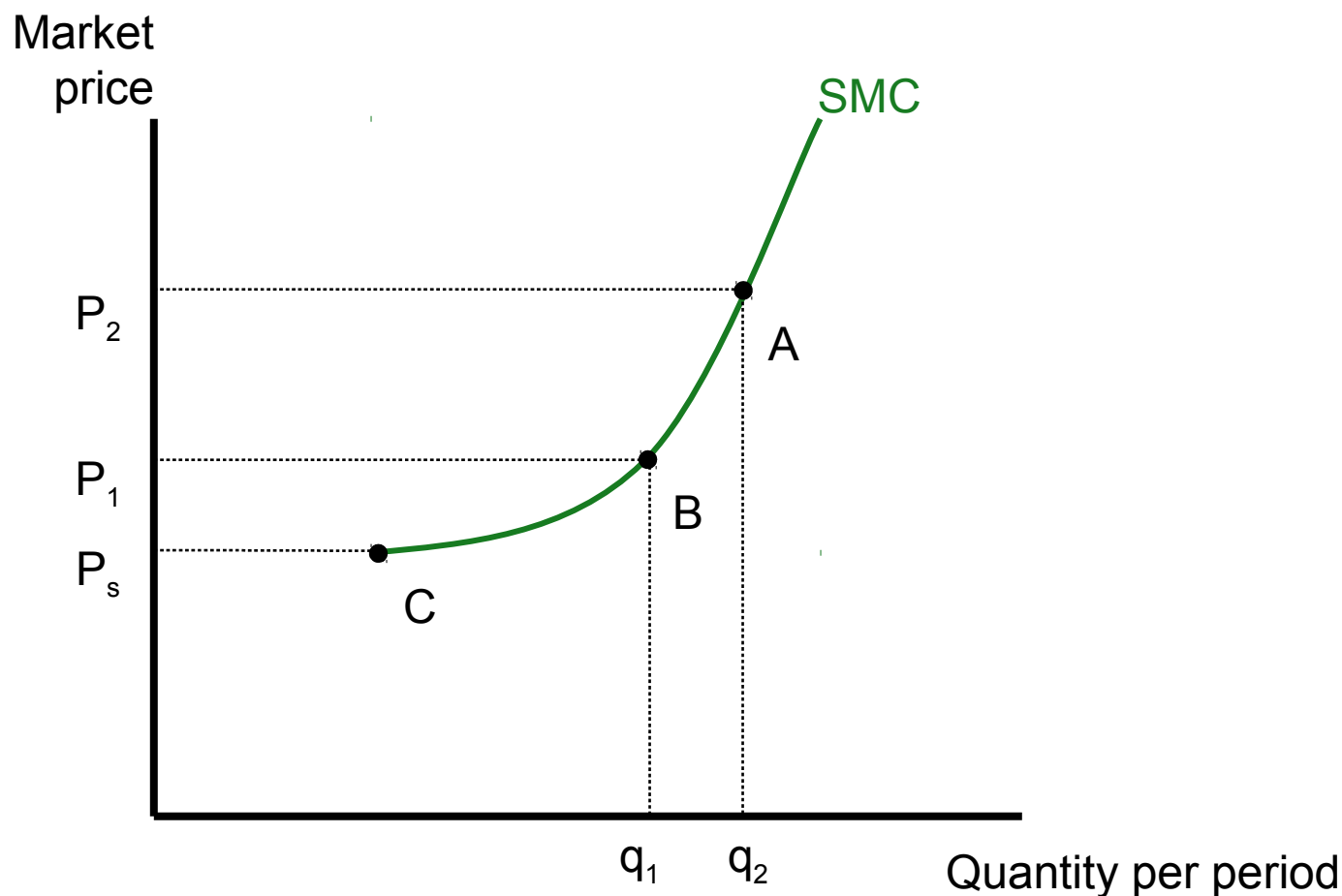
$$\frac{\partial \Pi(P, v, w)}{\partial v} = -k(P, v, w)$$

$$\frac{\partial \Pi(P, v, w)}{\partial w} = -l(P, v, w)$$

Producer Surplus in the Short Run

- Profit function is nondecreasing in output prices
 - If $P_2 > P_1$, $\Pi(P_2, \dots) \geq \Pi(P_1, \dots)$
 - The welfare gain to the firm of from the price change:
$$\text{welfare gain} = \Pi(P_2, \dots) - \Pi(P_1, \dots)$$

•Changes in Short-Run Producer Surplus Measure Firm Profits



•If price increases from P_1 to P_2 , then the increase in the firm's profits is given by area P_2ABP_1 . At a price of P_1 , the firm earns short-run producer surplus given by area P_sCBP_1 . This measures the increase in short-run profits for the firm when it produces q_1 rather than shutting down when price is P_s or below.

Producer Surplus in the Short Run

- Welfare gain:

$$\begin{aligned}\text{welfare gain} &= \Pi(P_2, \dots) - \Pi(P_1, \dots) = \\ &= \int_{P_1}^{P_2} \frac{\partial \Pi}{\partial P} dP = \int_{P_1}^{P_2} q(P) dP\end{aligned}$$

- Producer surplus

– The extra profits available from facing a price of P_1

$$\text{producer surplus} = \Pi(P_1, \dots) - \Pi(P_s, \dots) = \int_{P_s}^{P_1} q(P) dP$$

Producer Surplus in the Short Run

- **Producer surplus**
 - The extra return that producers make by making transactions at the market price
 - Over and above what they would earn if nothing were produced
 - The area below the market price and above the supply curve

Producer Surplus in the Short Run

- At the shutdown price
 - The firm produces no output, $\Pi(P_0, \dots) = -vk_1$
 - Profits = Fixed costs
 - Producer surplus = current profits + short-run fixed costs

$$\begin{aligned}\text{producer surplus} &= \Pi(P_1, \dots) - \Pi(P_0, \dots) = \\ &= \Pi(P_1, \dots) - (-vk_1) = \Pi(P_1, \dots) + vk_1 = \\ &= P_1 q_1 - wl_1\end{aligned}$$

11.4 A Short-Run Profit Function

- Cobb–Douglas production function, $q = k^\alpha l^\beta$
 - With $k = k_1$ in the short-run
- To find the profit function
 - Use the first-order conditions for a maximum

$$\frac{\partial \pi}{\partial l} = \beta P k_1^\alpha l^{\beta-1} - w = 0, \quad \text{so} \quad l = \frac{w}{\beta P k_1^\alpha} / (\beta - 1)$$

with $A = (w / \beta P k_1^\alpha)$

$$\Pi(P, v, w, k_1) = \frac{1 - \beta}{\beta^{\beta/(\beta-1)}} w^{\beta/(\beta-1)} p^{1/(1-\beta)} k_1^{\alpha/(1-\beta)} - v k_1$$

Profit Maximization and Input Demand

- A firm's output
 - Is determined by the amount of inputs it chooses to employ
- Relationship between inputs and outputs
 - Summarized by the production function
- A firm's economic profit
 - Can be expressed as a function of inputs

$$\pi(k, l) = Pq - C(q) = Pf(k, l) - (vk + wl)$$

Profit Maximization and Input Demand

- The first-order conditions for a maximum:

$$\partial\pi/\partial k = P[\partial f/\partial k] - v = 0$$

$$\partial\pi/\partial l = P[\partial f/\partial l] - w = 0$$

– Also imply cost minimization: $RTS = w/v$

- A profit-maximizing firm

– Should hire any input up to the point at which

- Its marginal contribution to revenues is equal to the marginal cost of hiring the input

Profit Maximization and Input Demand

- Marginal revenue product
 - The extra revenue a firm receives
 - When it uses one more unit of an input
 - In the price-taking case, $MRP_l = Pf_l$ and $MRP_k = Pf_k$

Profit Maximization and Input Demand

- Second-order conditions:

$$\pi_{kk} = f_{kk} < 0$$

$$\pi_{ll} = f_{ll} < 0$$

$$\pi_{kk}\pi_{ll} - \pi_{kl}^2 = f_{kk}f_{ll} - f_{kl}^2 > 0$$

- Capital and labor must exhibit sufficiently diminishing marginal productivities so that marginal costs rise as output expands

Input Demand Functions

- Input demand functions

$$\text{Capital Demand} = k(P, v, w)$$

$$\text{Labor Demand} = l(P, v, w)$$

- Are unconditional
- They implicitly allow the firm to adjust its output to changing prices

Single-Input Case

- We expect $\partial l / \partial w \leq 0$
 - Diminishing marginal productivity of labor
- The first order condition for profit maximization was: $Pf_l - w = 0$
 - Finding the derivative:
 - Use the Chain Rule!

Two-Input Case

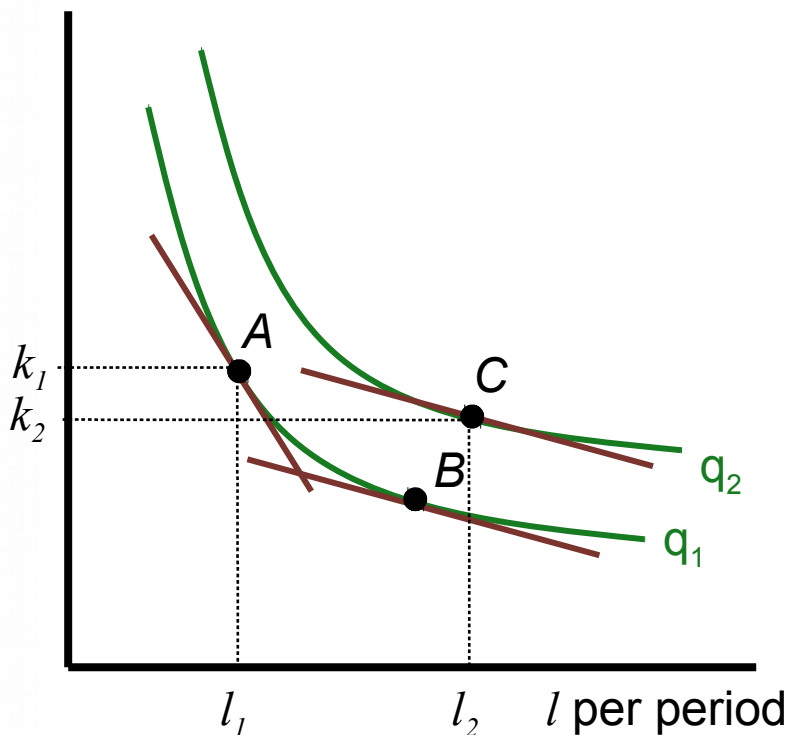
- For the case of two (or more inputs), the story is more complex
 - If there is a decrease in w , there will not only be a change in l but also a change in k as a new cost-minimizing combination of inputs is chosen
 - When k changes, the entire f_l function changes
- But, even in this case, $\partial l / \partial w \leq 0$

Two-Input Case

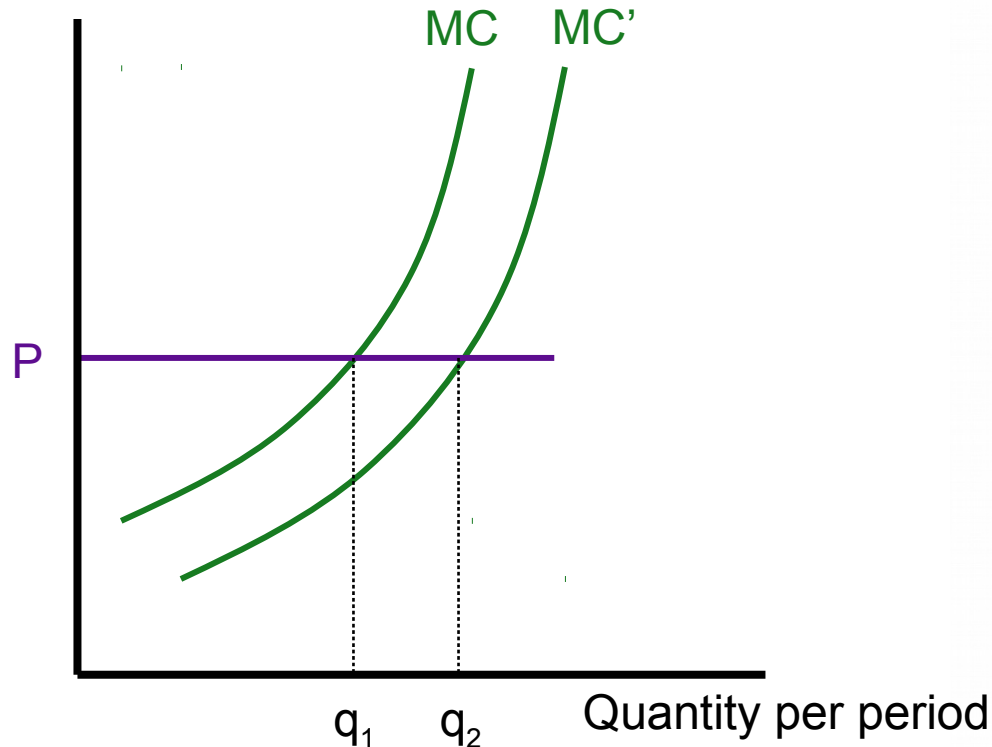
- When w falls
 - Substitution effect
 - If output is held constant, there will be a tendency for the firm to want to substitute l for k in the production process
 - Output effect
 - A change in w will shift the firm's expansion path
 - The firm's cost curves will shift and a different output level will be chosen

•The Substitution and Output Effects of a Decrease in the Price of a Factor

k per period (a) The isoquant map



Price (b) The output decision



•When the price of labor falls, two analytically different effects come into play. One of these, the substitution effect, would cause more labor to be purchased if output were held constant. This is shown as a movement from point A to point B in (a). At point B, the cost-minimizing condition ($RTS = w/v$) is satisfied for the new, lower w . This change in w/v will also shift the firm's expansion path and its marginal cost curve. A normal situation might be for the MC curve to shift downward in response to a decrease in w as shown in (b). With this new curve (MC') a higher level of output (q_2) will be chosen. Consequently, the hiring of labor will increase (to l_2), also from this output effect.

Cross-Price Effects

- How capital usage responds to a wage change
 - No definite statement can be made
 - A fall in the wage will lead the firm to substitute away from capital
 - The output effect will cause more capital to be demanded as the firm expands production

Substitution and Output Effects

- When the price of an input falls
 - Two effects cause the quantity demanded of that input to rise:
 1. The substitution effect causes any given output level to be produced using more of the input
 2. The fall in costs causes more of the good to be sold, thereby creating an additional output effect that increases demand for the input

Substitution and Output Effects

- Two concepts of demand for any input
 - Conditional demand for labor, $l^c(v, w, q)$
 - Unconditional demand for labor, $l(P, v, w)$
 - At the profit-maximizing level of output

$$l(P, v, w) = l^c(v, w, q) = l^c(v, w, q(P, v, w))$$

Substitution and Output Effects

- Differentiation with respect to w yields

$$\frac{\partial l(P, v, w)}{\partial w} = \underbrace{\frac{\partial l^c(v, w, q)}{\partial w}}_{\text{substitution effect}} + \underbrace{\frac{\partial l^c(v, w, q)}{\partial q} \cdot \frac{\partial q}{\partial w}}_{\text{output effect}}$$

total effect